

# A NOTE ON A toy model of two countries

## Content of the Handout

|   |                                      |      |
|---|--------------------------------------|------|
| 1 | Overview .....                       | p. 1 |
| 2 | Determinacy of the equilibrium ..... | p. 5 |

## 1. Overview

Let us suppose that there is two economies  $i = \{H, F\}$  of relative size  $n$  and  $1 - n$  respectively.

### 1.1 Households

Suppose a welfare maximizing household enjoying current and future consumption and dislike working with respective curvatures  $\sigma_C, \sigma_H > 0$ . The household maximizes the following stream of utilities discounted at a fixed rate  $\beta$ :

$$\max_{\{C_t, d_t, b_t, H_t\}} \sum_{\tau=0}^{\infty} \beta^\tau E_t \left\{ \frac{(C_{jt+\tau} - X_{t+\tau})^{1-\sigma_C}}{1-\sigma_C} - \chi_i \frac{H_{jt+\tau}^{1+\sigma_H}}{1+\sigma_H} \right\} \quad (1)$$

where  $C_{i,t}$  is consumption,  $X_{i,t}$  is the stock of external habits  $X_{i,t} = hC_{i,t-1}$  with  $h \in [0; 1)$  the degree of persistence habits,  $H_{i,t}$  is hours worked and  $\chi_i > 0$  is a shift parameter allowing to pin down the amount of hours worked in steady state.

The budget constraint in real terms reads as:

$$w_t H_{jt} + \frac{\Pi_{jt}}{P_t^C} + \frac{r_{t-1}}{\pi_t^C} d_{jt-1} + e_t \frac{r_{t-1}^*}{\pi_t^C} b_{jt-1} = C_{jt} + b_{jt} + e_t b_{jt}^* + e_t \frac{P_t}{P_t^C} g(b_{jt}), \quad (2)$$

where  $d_{jt}$  denotes the acquisition of domestic riskless bonds and  $b_{jt}$  foreign ones adjusted by the exchange rate  $e_t$ . Households pay an cost on foreign bonds, this is deemed necessary to remove an unit root component in consumption dynamics in an open economy environment. Here, the cost function reads as  $g(b_{jt}) = 0.5\chi_B (b_{jt} - \bar{b}_j)^2$ .

The Lagrangian problem reads as:

$$\begin{aligned}\mathcal{L}_t^H = & \sum_{\tau=0}^{\infty} \beta^\tau \left[ \frac{(C_{jt+\tau} - X_{t+\tau})^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+\tau}^{1+\sigma_H}}{1+\sigma_H} \right] \\ & + \beta^\tau \lambda_{t+\tau}^c \left[ w_t H_{jt} + \frac{\Pi_{jt}}{P_t^C} + \frac{r_{t-1}}{\pi_t^C} d_{jt-1} + e_t \frac{r_{t-1}^*}{\pi_t^C} b_{jt-1} \right] \\ & - \beta^\tau \lambda_{t+\tau}^c \left[ C_{jt} + d_{jt} + e_t b_{jt} + \frac{P_t}{P_t^C} e_t g(b_{jt}) \right]\end{aligned}$$

where  $\lambda_t^c$  is a lagrangian multiplier.

First order conditions are:

$$C_{jt} : (C_{jt} - X_t)^{-\sigma_C} - \lambda_t^c = 0 \quad (3)$$

$$d_{jt} : \beta E_t \{ \lambda_{t+1} \} r_t / E_t \{ \pi_{t+1}^C \} - \lambda_t^c = 0 \quad (4)$$

$$b_{jt} : \beta E_t \{ \lambda_{t+1} e_{t+1} \} r_t^* / E_t \{ \pi_{t+1}^C \} - e_t \lambda_t^c = 0 \quad (5)$$

$$H_{jt} : -\chi H_{jt}^{\sigma_H} + \lambda_t^c w_t = 0 \quad (6)$$

We define the real exchange rate as:

$$rer_t = e_t \frac{P_t^{C*}}{P_t^C}. \quad (7)$$

The households allocates their consumption on domestically produced goods and also imports foreign goods. The consumption basket of the household:

$$C_{jt} = \left( (1 - \alpha_C)^{\frac{1}{\mu}} (c_{jt})^{\frac{\mu-1}{\mu}} + (\varepsilon_t^X \alpha_C)^{\frac{1}{\mu}} (c_{jt}^*)^{\frac{\mu-1}{\mu}} \right)^{\frac{\mu}{\mu-1}} \quad (8)$$

where  $\alpha_C \in [0, 0.5]$  is the fraction of foreign goods in the consumption basket, conversely  $1-\alpha_C$  is referred to as the home bias. In addition,  $\varepsilon_t^X$  is an AR(1) shock process of the demand for foreign goods and can be interpreted as an import shock. Parameter  $\mu \geq 0$  is the substitution degree between home goods and foreign goods, when  $\mu$  is high, households are more sensible to price differential between home and foreign goods, and will reduce their imports.

The price index of goods reads as:

$$P_t^C = \left[ (1 - \alpha_C) (P_t)^{1-\mu} + \alpha_C (e_t P_t^*)^{1-\mu} \right]^{\frac{1}{1-\mu}} \quad (9)$$

Optimal demands for home  $c_{jt}$  and foreign  $c_{jt}^*$  varieties are given by:

$$c_{jt} = (1 - \alpha_C) \left( \frac{P_t}{P_t^C} \right)^{-\mu} C_{jt} \text{ and } c_{jt}^* = \varepsilon_t^X \alpha_C \left( \frac{e_t P_t^*}{P_t^C} \right)^{-\mu} C_{jt} \quad (10)$$

Here,  $c_{jt}^*$  are imports.

## 1.2 Retailers

Final firms competitively pack different types of good varieties  $i$  drawn on an unitary continuum to produce an homogeneous good sold to households. This firm is necessary when introducing monopolistic

competition as they allows to create an elastic demand curve that intermediate firms exploits to generate positive profits. Production of final firms employs the following CES packing technology:

$$Y_t = \left( \int_0^1 y_{it}^{(\epsilon-1)/\epsilon} di \right)^{\epsilon/(\epsilon-1)} \quad (11)$$

where  $\epsilon > 1$  is the perfect substitutability between different types of varieties  $i$ . The economy converges to a competitive equilibrium for  $\epsilon \rightarrow 1$ , while for any  $\epsilon > 1$ , goods are imperfect substitutes which allows firms to rise their selling price above their marginal cost.

Final firms are perfectly competitive and maximize profits,  $P_t Y_t - p_{it} y_{it}$ , under their technology constraint in Equation 11 to obtain the optimal demand constraint for the  $i$ -th good:

$$y_{it} = \left( \frac{p_{it}}{P_t} \right)^{-\epsilon} Y_t. \quad (12)$$

While the zero profit condition reads as:  $P_t = (\int_0^1 p_{it}^{1-\epsilon} di)^{1/(1-\epsilon)}$ .

### 1.3 Intermediate firms

Let us now suppose that there is a large number of identical firms. They hire workers and combine them with capital (both produced by them and supplied by households). Intermediate firms' technology is given by:

$$Y_{it} = \varepsilon_t^Z H_{it}^\alpha \quad (13)$$

Profits of the intermediate firm  $i$  are given by:

$$\Pi_{it} = P_{it} \varepsilon_t^Z H_{it}^\alpha - W_t H_{it} \quad (14)$$

Intermediate firms solve a two stage problem. First they minimize input costs on a perfectly competitive market. First order condition provides the marginal cost:

$$MC_{it} = \frac{H_{it}}{\alpha Y_{it}} W_t \quad (15)$$

Second, firms face price adjustment costs paid in terms of home goods  $P_t$ . These are quadratic adjustment costs:

$$\Delta_{it}^P = \frac{\chi_P}{2} \left( \frac{P_{it+\tau}}{P_{it-1+\tau}} - 1 \right)^2 \quad (16)$$

The profit optimization becomes dynamic, intermediates set prices facing nominal rigidities by maximizing the discount sum of real profits:

$$\max_{p_{it}} \sum_{\tau=0}^{\infty} \beta^\tau \frac{\lambda_{t+\tau}^c}{\lambda_t^c} \left[ \frac{P_{it+\tau}}{P_{t+\tau}^C} \left( \frac{P_{it+\tau}}{P_{t+\tau}} \right)^{-\epsilon} - \varepsilon_{t+\tau}^P mc_{t+\tau} \left( \frac{P_{it+\tau}}{P_{t+\tau}} \right)^{-\epsilon} - \frac{P_{t+\tau}}{P_{t+\tau}^C} \frac{\chi_P}{2} \left( \frac{P_{it+\tau}}{P_{it-1+\tau}} - 1 \right)^2 \right] Y_{t+\tau} \quad (17)$$

where  $mc_t$  is the marginal cost and  $\varepsilon_t^P$  is an AR(1) price shock.

First order condition, assuming symmetry (i.e.  $P_t = P_{it}$ ) :

$$(1 - \epsilon) + \epsilon \varepsilon_t^P mc_t - \chi_P \pi_t (\pi_t - 1) + \beta \frac{\lambda_{t+1}^c}{\lambda_t^c} \frac{Y_{t+1}}{Y_t} \chi_P \pi_{t+1} (\pi_{t+1} - 1) = 0 \quad (18)$$

## 1.4 Authorities

We use a ? rule-type to model the setting of the nominal interest rate by the monetary authority:

$$R_t = R_{t-1}^\rho \left( \bar{R} \left( \frac{\pi_t^C}{\bar{\pi}^C} \right)^{\phi_\pi} \left( \frac{Y_t}{\bar{Y}} \right)^{\phi_y} \right)^{1-\rho} \varepsilon_t^R \quad (19)$$

where  $\rho \in [0, 1)$  is the smoothing coefficient of the monetary policy rule,  $\phi_\pi$  the reaction to the inflation gap and  $\phi_y$  the reaction to the output gap. Steady states for interest rates, inflation and output are respectively  $\bar{R}$ ,  $\bar{\pi}$  and  $\bar{Y}$ . Finally,  $\varepsilon_t^R$  is a monetary policy shock.

## 1.5 Equilibrium conditions

First, the nominal exchange rate and real exchange rates in both sides of each countries are related by:

$$e_t = 1/e_t^* \quad \text{and} \quad rer_t = 1/rer_t^*$$

we can thus substitute for the foreign country and express our model in terms of  $e_t$  and  $rer_t$ .

Resource constraints is given by the total demand for home goods from home and foreign households (including also adjustment costs)

$$\int_0^n Y_{it} di = \int_0^n c_{jt} dj + \int_n^1 c_{jt}^* dj + \int_0^n \Delta_t^P Y_{jt} dj + \int_0^n e_t g(b_t) dj \quad (20)$$

$$nY_t = n(1 - \alpha_C) \left( \frac{P_t}{P_t^C} \right)^{-\mu_C} C_t + (1 - n) \varepsilon_t^{X*} \alpha_C \left( \frac{P_t}{e_t P_t^{C*}} \right)^{-\mu_C} C_t^* + n \Delta_t^P Y_t + n g(b_t) \quad (21)$$

$$Y_t = (1 - \alpha_C) \left( \frac{P_t}{P_t^C} \right)^{-\mu_C} C_t + \frac{(1 - n)}{n} \varepsilon_t^{X*} \alpha_C \left( \frac{P_t}{e_t P_t^{C*}} \right)^{-\mu_C} C_t^* + \Delta_t^P Y_t + \frac{\chi_B}{2} (b_t - \bar{b})^2 \quad (22)$$

Similarly for the foreign country:

$$\int_n^1 Y_{it}^* di = \int_n^1 c_{jt}^* dj + \int_1^n c_{jt} dj + \int_n^1 \Delta_t^P Y_{jt} dj + \int_n^1 \frac{\chi_B}{2} (b_{jt}^* - \bar{b}_j)^2 dj$$

To get the accumulation of net foreign asset, we aggregate the household constraint and assume the amount of domestic bonds is zero,  $B_t = B_{t-1} = 0$ :

$$e_t n B_t - e_t R_{t-1}^* n B_{t-1} = W_t n H_t + n \Pi_t - P_t^C n C_t - P_t n g(b_t)$$

Recall that with a zero profit condition over inputs markets, we have,  $MC_t Y_t = W_t H_t$ . While on final goods, a markup is applied net of adjustment costs of prices,  $\Pi_t = [P_t - MC_t - \Delta_t^P] Y_t$ . We find after rewriting and injecting  $Y_t$ :

$$\begin{aligned} e_t B_t - e_t R_{t-1}^* B_{t-1} &= MC_t Y_t + [P_t - MC_t - P_t \Delta_t^P] Y_t - P_t^C C_t - P_t g(b_t) \\ &= P_t Y_t - P_t \Delta_t^P Y_t - P_t^C C_t - P_t g(b_t) \\ &= P_t \left[ (1 - \alpha_C) \left( \frac{P_t}{P_t^C} \right)^{-\mu_C} C_t + \varepsilon_t^{X*} \alpha_C \left( \frac{P_t}{e_t P_t^{C*}} \right)^{-\mu_C} C_t^* \frac{1 - n}{n} \right] - P_t^C C_t \end{aligned}$$

This equation determines the quantity of foreign asset at the end of period, this way the left hand side of the equation needs to be rewritten in order to take into account the timing effect of the exchange rate over the previous stock of foreign debt. In particular, an increase in the foreign exchange rate  $e_t/e_{t-1} > 1$  generates

## 2. Determinacy of the equilibrium

---

a wealth effect for the domestic household by boosting his financial revenues. The left hand side including this end of period wealth effect from changes in the nominal exchange rate reads as,  $B_t^* - e_t/e_{t-1}R_{t-1}^*B_{t-1}^*$ . In real terms, the final equation of net foreign assets is given by:

$$b_{jt} = \frac{R_{t-1}^*}{\pi_t^C} \Delta e_t b_{jt-1} + \frac{P_t}{P_t^C} \left( (1 - \alpha_C) \left( \frac{P_t}{P_t^C} \right)^{-\mu_C} C_t + \varepsilon_t^{X^*} \alpha_C \left( \frac{P_t}{e_t P_t^*} \right)^{-\mu_C} C_t^* \frac{1-n}{n} \right) - C_t \quad (23)$$

In a world populated by two countries, international financial markets are closed by:

$$nb_t + (1 - n) b_t^* = 0 \quad (24)$$

## 2. Determinacy of the equilibrium

---

For determinacy purpose, some variables need to be rewritten to be stationnary. In particular, prices and the exchange rate are not stationnary, thus requiring to express (i) the nominal exchange rate as variations  $\Delta e_t$ , and (ii) prices in a relative fashion  $p_t = P_t/P_t^C$ .

First, the relative price index can be expressed in growth rates to be stationnary:

$$\frac{p_t}{p_{t-1}} = \frac{\pi_t}{\pi_t^C} \quad (25)$$

Second, recall that  $rer_t = e_t P_t^{C*}/P_t^C$ , then we can express the real exchange in terms of growth too:

$$\frac{rer_t}{rer_{t-1}} = \Delta e_t \frac{\pi_t^{C*}}{\pi_t^C} \quad (26)$$

And the FOC on foreign bonds can also be written in terms of variations of the nominal exchange rate:

$$\mathbb{E}_t \{ \Delta e_{t+1} \} = \left( 1 + \chi_B rer_t (b_t - \bar{b}) \right) \frac{R_t}{R_t^*} \quad (27)$$

Third, dividing the CES price index by  $P_t^C$ , the following relation emerges:

$$1 = \left[ (1 - \alpha) p_t^{1-\mu} + \alpha rer_t^{1-\mu_C} \right]^{\frac{1}{1-\mu}} \quad (28)$$

In addition, relative price setting affect the following equations:

$$Y_t = (1 - \alpha_C) p_t^{-\mu} C_t + \frac{1-n}{n} \alpha_C \left( \frac{p_t}{rer_t} \right)^{-\mu} C_t^* + 0.5 \chi_P (\pi_t - 1)^2 Y_t + g(b_t) \quad (29)$$

$$b_t^* = \frac{R_{t-1}^*}{\pi_t^C} \Delta e_t b_{t-1}^* + p_t \left( (1 - \alpha_C) (p_t)^{-\mu} C_t + \frac{1-n}{n} \varepsilon_t^{X^*} \alpha_C \left( \frac{p_t}{rer_t} \right)^{-\mu} C_t^* \right) - C_t \quad (30)$$

## References

---